USER INTERFACE DESIGN DOCUMENT

Autonomous Video Ad-Placement & Contextual Safety Analyzer

Submitted in partial fulfillment of the requirements for the course

Specialization Project

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1. Introduction

The purpose of this document is to describe the user interface design of the **Autonomous Video Ad-Placement & Contextual Safety Analyzer**, a locally hosted desktop application that audits long-form video for brand-safety risks and identifies premium advertisement placement surfaces. This document provides the screen layouts, navigation flow, interface standards, validation rules, and design decisions that govern the system's presentation layer.

The application is branded **AdSafe Studio** in the interface. Its visual design deliberately follows the conventions of professional creator tooling — a persistent top bar, a left navigation rail, and a content table with inline status — so that the target users (video creators and marketing teams) encounter an interface whose structure is already familiar to them.

All screen figures in this document are **live captures of the working application**, not mockups. The interface is fully implemented and operational.

2. User Roles

2.1 Usage Personas

The system is a single-user desktop tool. The following personas describe the *usage contexts* the interface is designed to serve, as identified in the Software Requirements Specification.

Persona	**Goals served by the interface**
Video Creator / Editor	Verify brand safety and monetization eligibility before publishing; locate sponsor placement opportunities
Digital Marketing Agency	Rapidly audit creator content before attaching a client brand; export evidence of the audit
Compliance Officer	Review flagged material against organisational policy; configure the restricted keyword dictionary and detection thresholds

2.2 Authentication: Deliberately Out of Scope

The application does **not** implement user accounts, login, or role-based access control, and no such screens exist. This is a considered design decision rather than an omission:

- The system is specified as an **offline, locally hosted** application. All media and audit data remain on the operator's own machine and are never transmitted.
- Introducing an authentication layer would add a credential store without adding any security benefit, since access to the application already implies access to the host filesystem on which its database and media reside.
- The privacy objective in the SRS is satisfied by *data never leaving the machine*, not by access control within it.

Accordingly the personas above are not authenticated roles; every persona operates the same single-user interface. Should the system later be deployed as a shared or multi-tenant service, authentication and role separation would be introduced at that point.

3. Navigation Structure

The application is a single-page application with client-side routing. A persistent application shell (top bar and navigation rail) frames every screen, so the primary destinations are reachable in one click from anywhere. The Upload dialog is modal and can be invoked from any screen through the **Create** action.

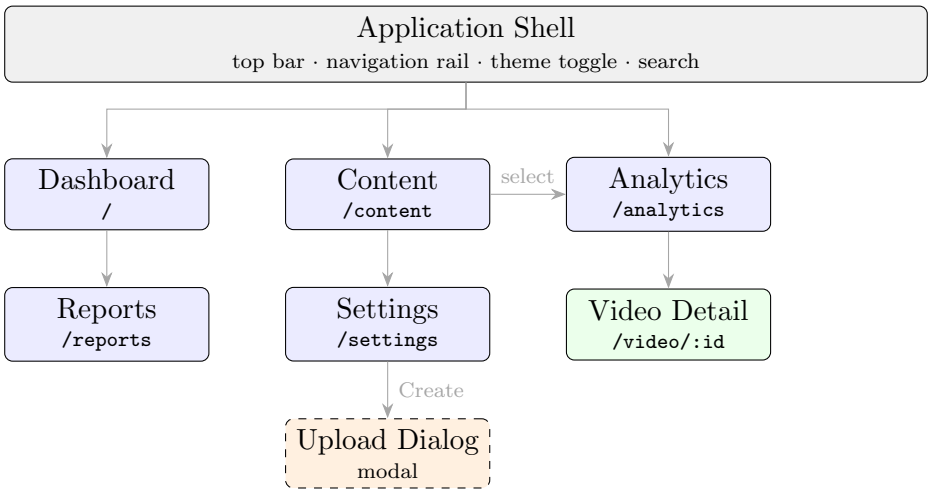


Figure 1: System Navigation Flow. Blue: primary destinations in the navigation rail. Green: contextual screen reached by selecting a video. Orange (dashed): modal overlay invokable from any screen.

Secondary navigation paths:

- Dashboard → Video Detail (via the *Recent uploads* list)
- Dashboard → Analytics (via *See full analytics*)
- Video Detail → Reports (report generated directly from the detail header)
- Any screen → Dashboard (via the application logo)

4. User Interface Standards

4.1 Typography

A single typeface is used throughout. Weight and size, not font family, carry the visual hierarchy.

Element	Font	Specification
Page title	Roboto Regular	25 px / weight 400
Section heading	Roboto Medium	15 px / weight 500
Body text	Roboto Regular	14 px / weight 400
Secondary text	Roboto Regular	12–13 px, muted colour
Button label	Roboto Medium	14 px / weight 500
Statistic value	Roboto Regular	32 px / weight 400
Icons	Material Symbols	Outlined, 20–24 px

4.2 Colour Palette

The interface ships a **dark theme by default** with a light theme available via a toggle in the top bar. Colour values were selected against measured contrast rather than by eye.

Surfaces and text

Role	Dark	Light	Contrast (dark)
Page background	`#0F0F0F`	`#F9F9F9`	—
Surface (cards)	`#212121`	`#FFFFFF`	—
Primary text	`#F1F1F1`	`#0F0F0F`	14.26:1
Secondary text	`#AAAAAA`	`#606060`	6.93:1
Action / primary	`#3EA6FF`	`#065FD4`	6.21:1
Accent (brand)	`#FF4E45`	`#CC0000`	4.93:1

Note on the action colour. The light-theme blue `#065FD4` measures only **2.76:1** against the dark card surface, which fails the WCAG 2.1 minimum of 3:1 for user-interface components. The dark theme therefore substitutes `#3EA6FF` (6.21:1). The action colour is a themed token rather than a fixed value for this reason.

Status palette (fixed across both themes)

Severity colours are held constant in both themes so that a given risk level always reads identically. Each is verified at ≥3:1 against both surfaces, and the worst adjacent pair separates by $\Delta E = 16.4$ under simulated deuteranopia (target ≥12), so the palette remains distinguishable to colour-blind users.

Meaning	Value	Applied to
High severity	`#D03B3B`	High-risk speech, weapon detections
Medium severity	`#EC835A`	Medium-risk speech, substances
Low severity	`#FAB219`	Low-risk speech
Positive / opportunity	`#0CA30C`	Ad spots, monetizable verdict
Limited monetization	`#F5A623`	Restricted verdict badge

Colour is never the sole carrier of meaning. Every severity indicator pairs its colour with a text label, and every status icon pairs with descriptive text.

4.3 Layout and Components

- **Grid:** fixed 240 px navigation rail; fluid content column capped at 1400 px.
- **Elevation:** a three-step surface ladder (page → surface → raised) separated by 1 px borders. Dark interfaces separate by border, not shadow.
- **Radius:** 12 px cards, 18 px pill buttons, 8 px inputs, 999 px chips.
- **Spacing:** 4 px base unit; 24 px between major blocks.
- **Feedback:** inline progress bars for long operations; transient snackbar confirmations; skeleton placeholders while loading.

5. Screen Design

Seven screens comprise the interface. Each is described by its purpose, its constituent components, and a capture of the implemented screen.

5.1 Dashboard Screen

Purpose

Landing screen. Summarises the state of the entire library so the operator can judge overall monetization health at a glance and resume recent work.

Components

- Four summary statistic cards — videos in library, fully monetizable count, total risk flags, total advertisement inventory
- Library health panel with a proportional monetizable / restricted split bar
- Recent uploads list with per-video status icons
- Shortcut links to Analytics and Content
- Empty state guiding a first-time user to upload

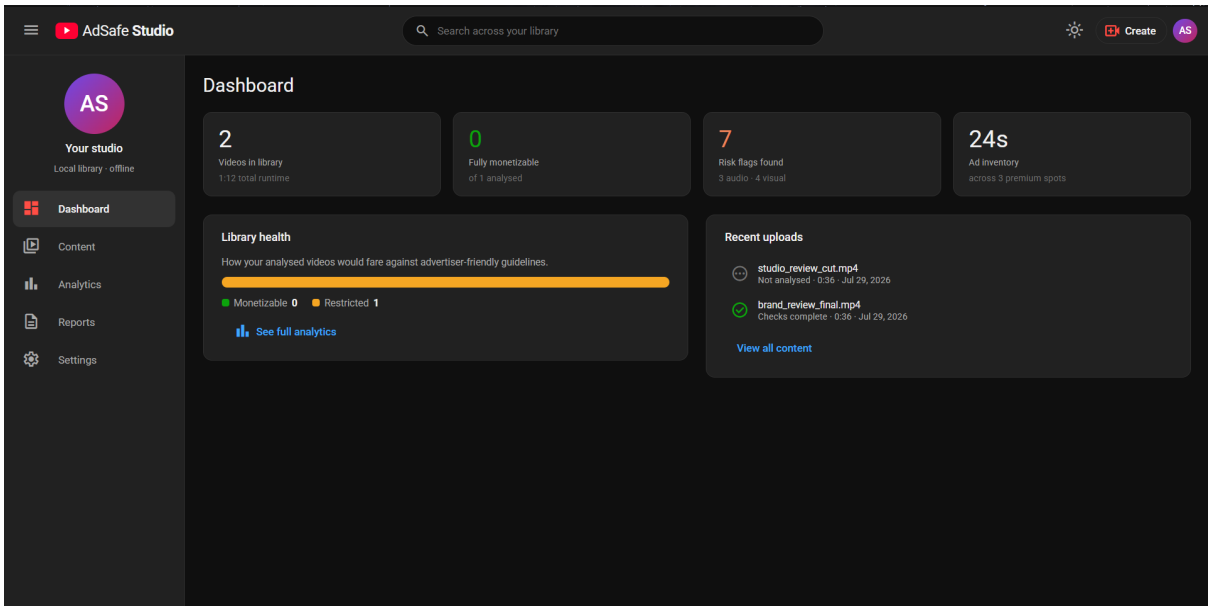


Figure 2: Dashboard Screen

5.2 Content Screen

Purpose

The video library. Lists every ingested video with its monetization verdict and analysis status, and is the launch point for running checks.

Components

- Sortable table: thumbnail, filename, technical metadata, monetization badge, status, upload date, duration, flag counts, advertisement spot count
- Monetization badge — green currency icon (monetizable), amber (limited), grey (not yet analysed)
- Per-row hover actions: open, run/re-run checks, delete
- Live progress bar with stage label while analysis runs
- Filter bar and library-wide search
- Empty state with a demo-video shortcut

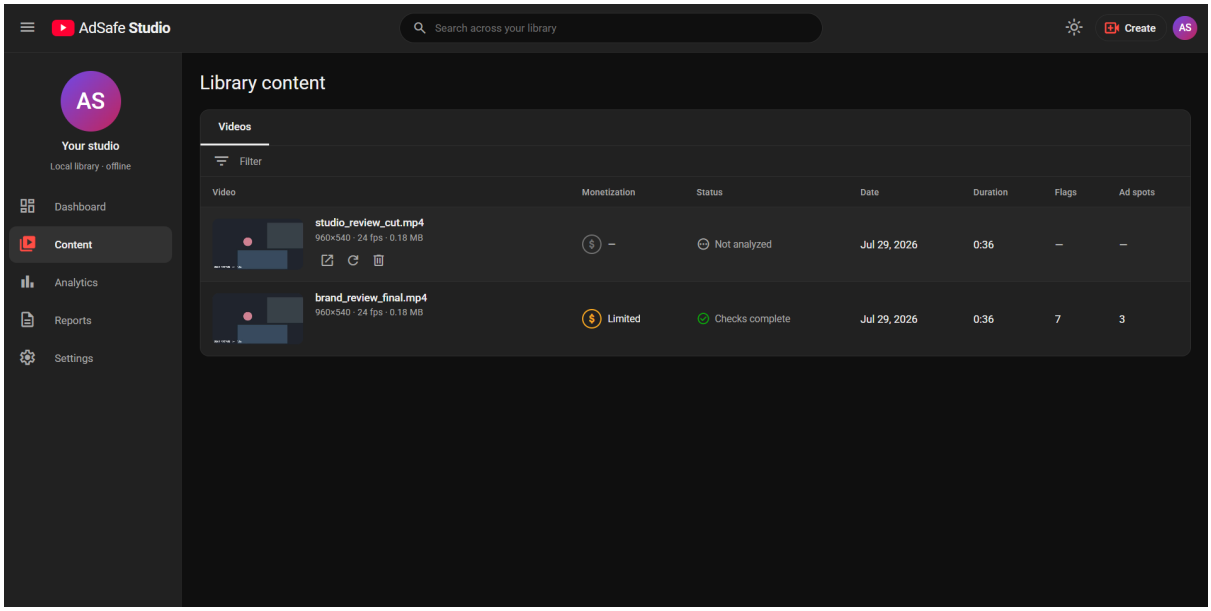


Figure 3: Content Screen

5.3 Video Detail Screen

Purpose

The primary analysis workspace. Presents every finding for a single video on a common timeline synchronised with video playback.

Components

- Header with filename, monetization verdict badge, and export actions
- Four statistic cards — audio flags, visual risks, premium advertisement spots, advertisement inventory
- **Audit timeline:** three lanes (audio, visual, advertisement spots) with colour-coded events, minute:second axis, and a live playhead. Events are clickable and keyboard-operable; selecting one seeks the player to that moment
- Embedded video player with native controls
- Tabbed result tables — audio flags (word, severity, transcript context), visual risks (object, category, confidence, bounding box), advertisement spots (interval, surface type, area, suitability)
- Metadata panel

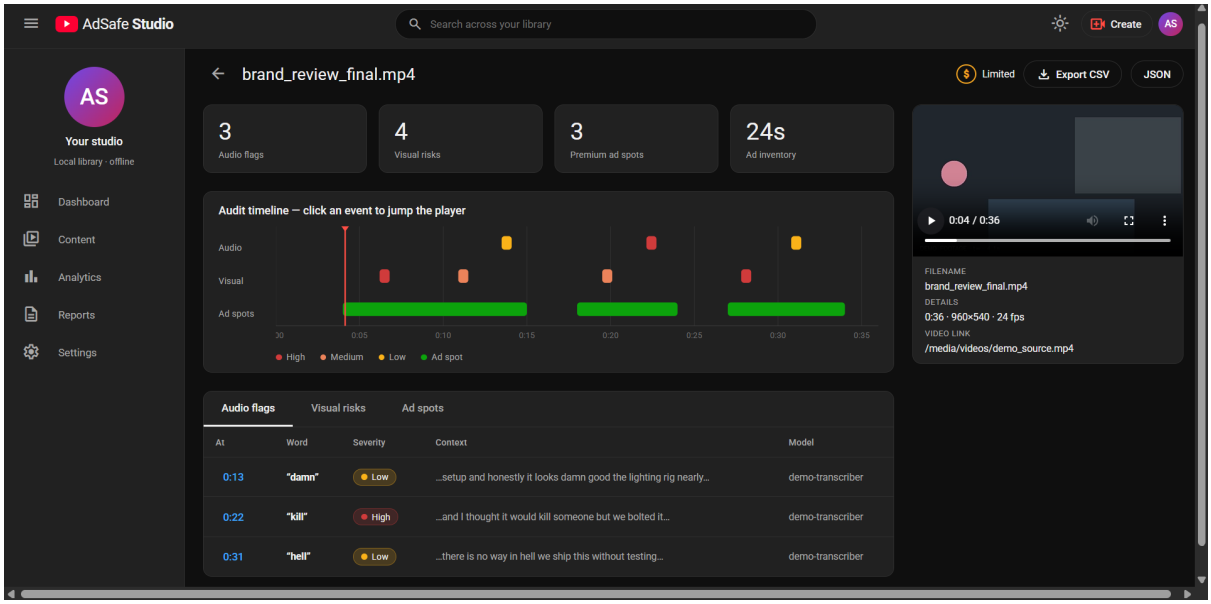


Figure 4: Video Detail Screen, showing the audit timeline synchronised with playback

5.4 Analytics Screen

Purpose

Aggregates findings across every analysed video, revealing recurring risks that are not visible when inspecting one video at a time.

Components

- Three summary statistic cards
- Monetization outcome split bar
- Audio flags by severity (status palette)
- Visual risks by category
- Most frequently flagged words
- Most frequently detected objects
- Advertisement inventory by surface type, in seconds

Each chart prints its numeric value beside every bar, so the visualisation also serves as its own data table.

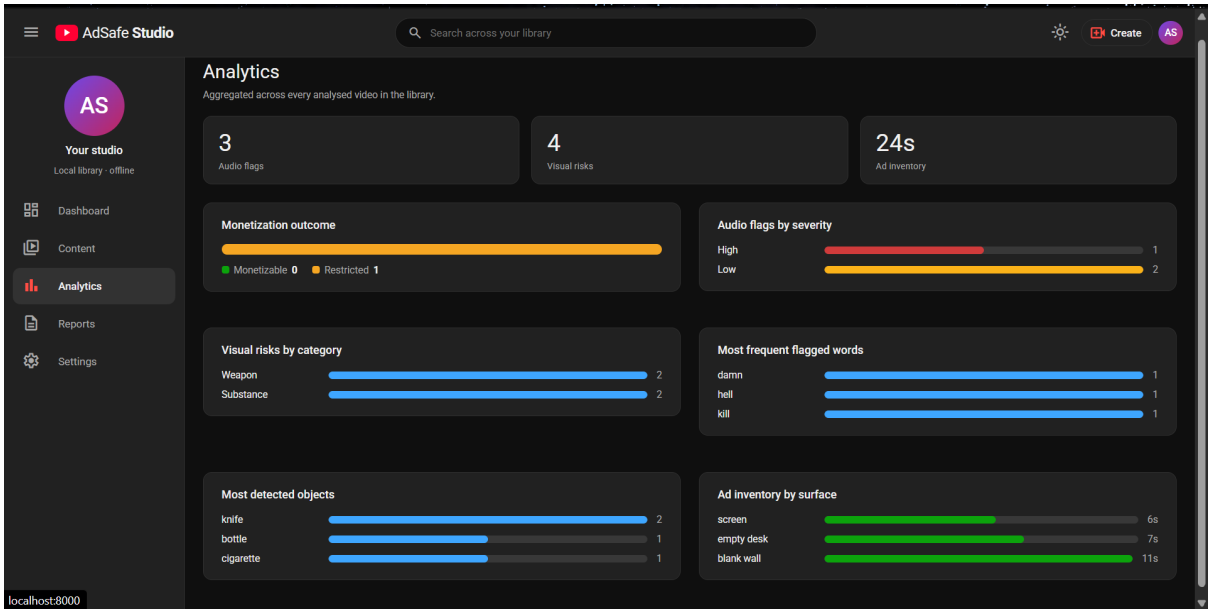


Figure 5: Analytics Screen

5.5 Reports Screen

Purpose

Generates and archives formal compliance reports for individual videos.

Components

- Video selector, restricted to analysed videos
- Output format selector — CSV, JSON
- Generate action
- Report history table — generation timestamp, source video, format, risk totals, compliance verdict
- Per-row download action

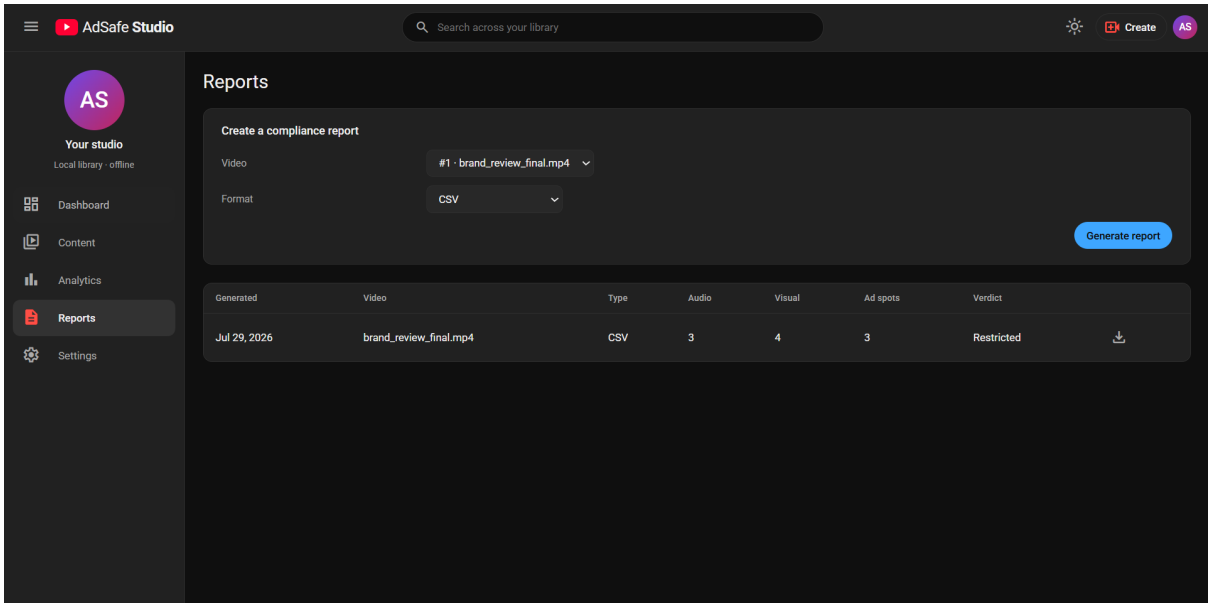


Figure 6: Reports Screen

5.6 Settings Screen

Purpose

Configures detection behaviour. Changes are persisted and applied to subsequent analysis runs.

Components

- Editable restricted keyword dictionary — term and severity per row, with add and remove actions
- Pipeline parameters — frame sample rate, minimum advertisement spot persistence, detection confidence threshold
- Transcription model selector
- Save action, disabled while any field is invalid
- Inline validation messages against the offending field

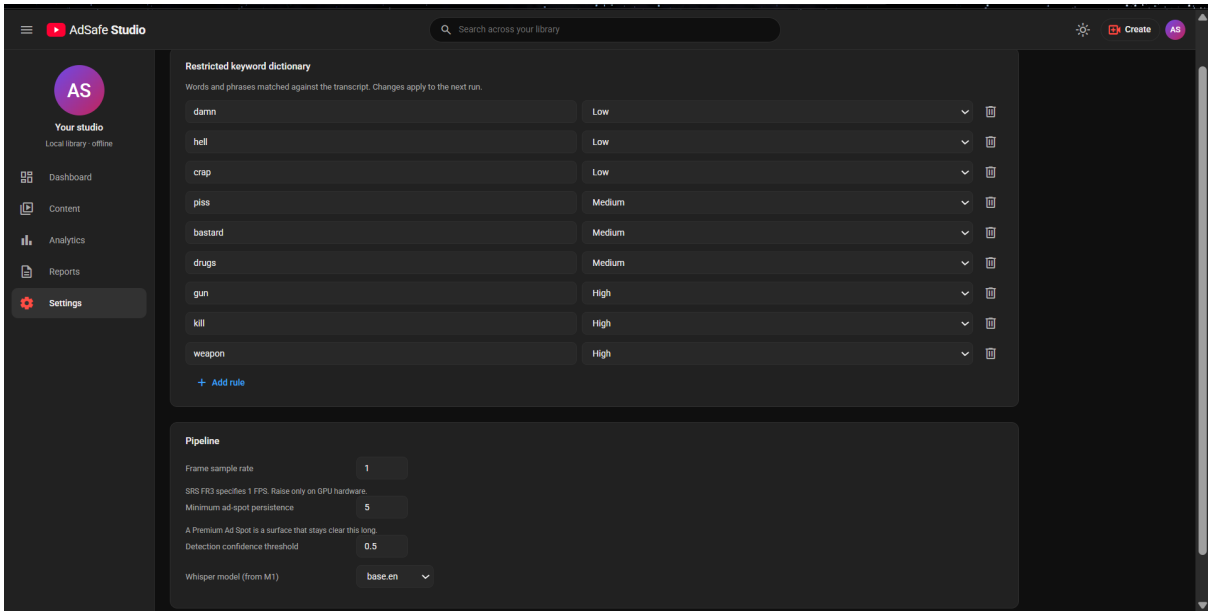


Figure 7: Settings Screen

5.7 Upload Dialog

Purpose

Modal overlay for ingesting a video. Reachable from any screen via the **Create** action.

Components

- Drag-and-drop zone with an equivalent file-picker button
- Accepted-format and size guidance
- Privacy reassurance stating that media never leaves the machine
- Determinate upload progress bar
- Inline validation message for rejected files
- Dismiss action, disabled during transfer to prevent interruption

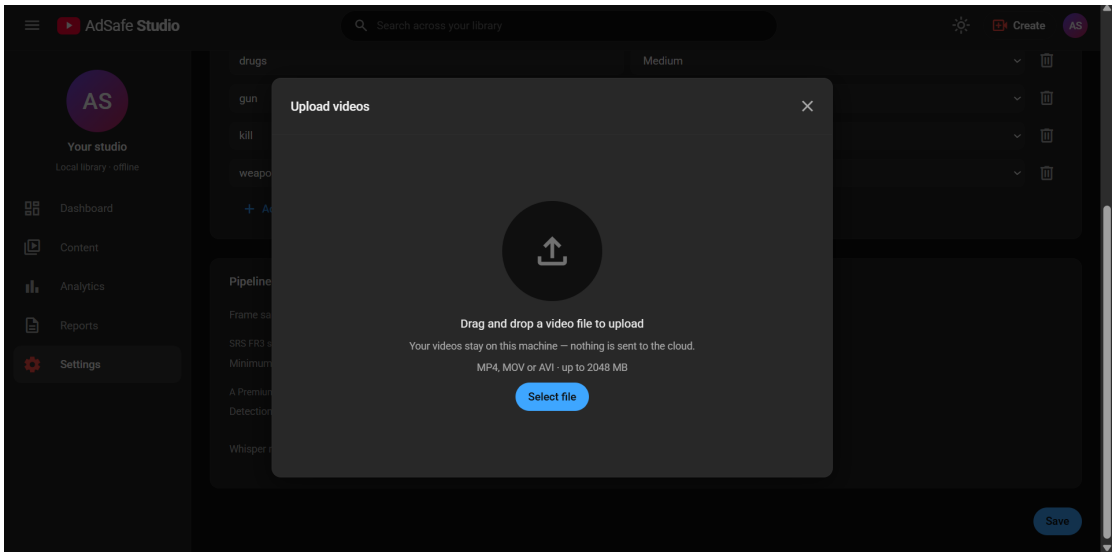


Figure 8: Upload Dialog

5.8 Interaction States

Two states warrant illustration because they define how the interface communicates progress and error.

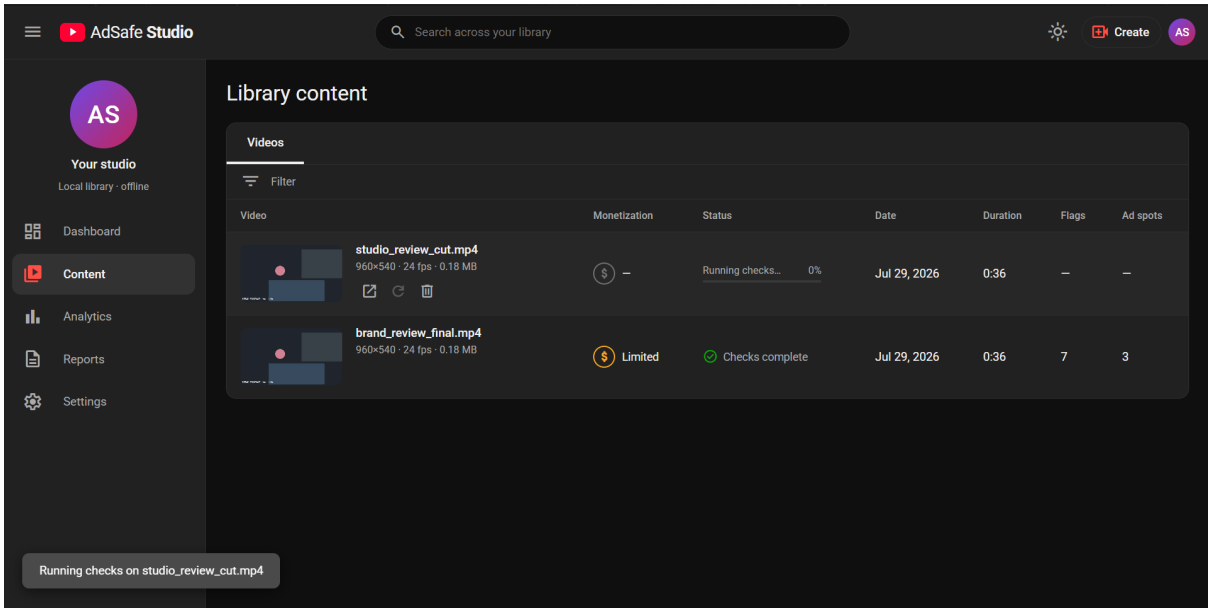


Figure 9: Analysis in progress. The status cell reports the active pipeline stage and completion percentage, updated continuously without a page reload.

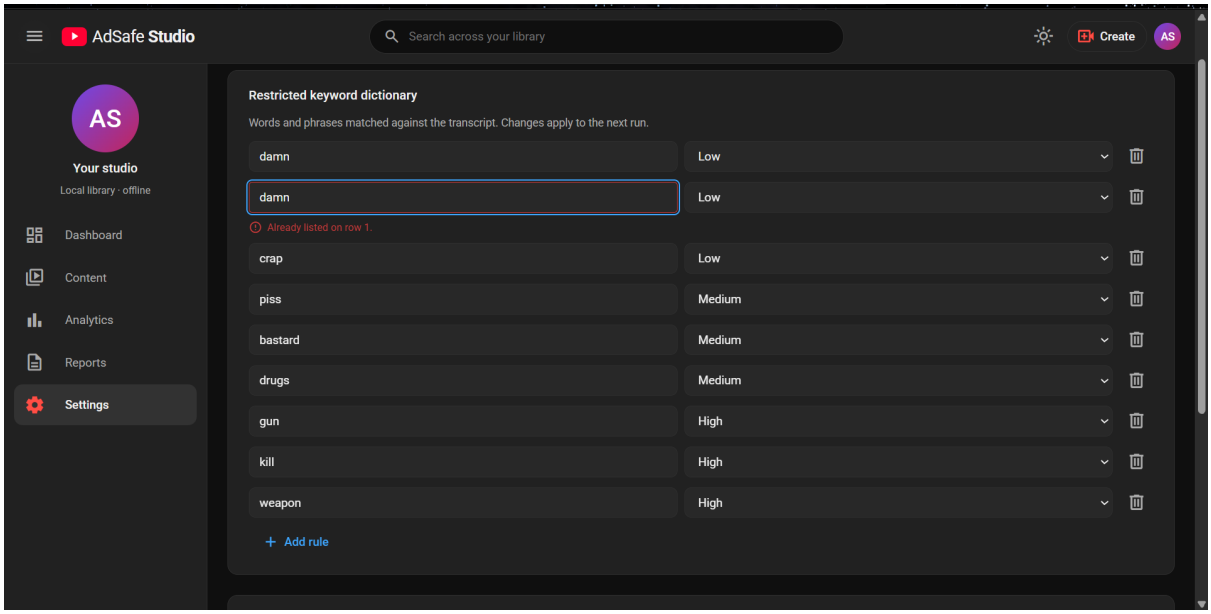


Figure 10: Inline validation. A duplicate dictionary entry is identified against the offending field, and the Save action is disabled until it is resolved.

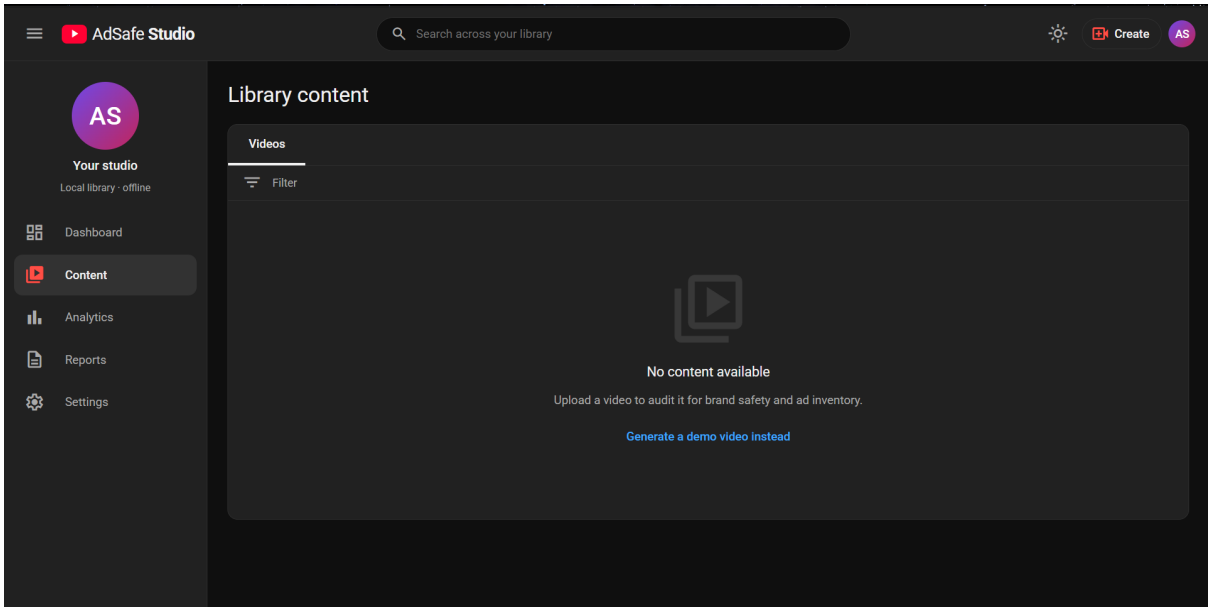


Figure 11: Empty state. Rather than presenting a blank table, the screen explains the situation and offers the next action.

6. Field Validation Rules

Validation is enforced in the interface before submission, and again on the server. Errors are reported inline, adjacent to the field responsible, and the corresponding submit action is disabled while any error is outstanding.

Field	Type	Validation Rule
Video file	File picker	Mandatory. Extension must be `.mp4`, `.mov` or `.avi`; file must be non-empty and at most 2048 MB; must be a decodable video stream
Restricted term	Text	Mandatory; leading and trailing whitespace trimmed; must be unique within the dictionary (case-insensitive)
Term severity	Dropdown	Mandatory; one of Low, Medium, High

Field	Type	Validation Rule
Frame sample rate	Number	Mandatory; numeric; range 0.25–4.0 frames per second
Minimum spot persistence	Number	Mandatory; numeric; range 2.0–15.0 seconds
Confidence threshold	Number	Mandatory; numeric; range 0.00–1.00
Transcription model	Dropdown	Mandatory; one of the supported model sizes
Report source video	Dropdown	Mandatory; restricted to videos whose analysis has completed
Report format	Dropdown	Mandatory; one of CSV, JSON
Library search	Text	Optional; free text; filters the table as typed

7. Responsive Design

The interface adapts across four breakpoints. The layout is fluid between them; the breakpoints mark where the structure itself changes.

View	Width	Adaptation
Desktop	$\geq$1200 px	Full navigation rail; two-column detail layout; four statistic cards per row
Laptop	900–1199 px	Navigation collapses to an icon rail; detail layout becomes single column; statistics two per row
Tablet	600–899 px	Navigation becomes an overlay drawer with a dimming scrim, opened from the menu control; content spans full width
Mobile	<600 px	Data table re-renders as a card list; search collapses; actions become icon-only; statistics two per row

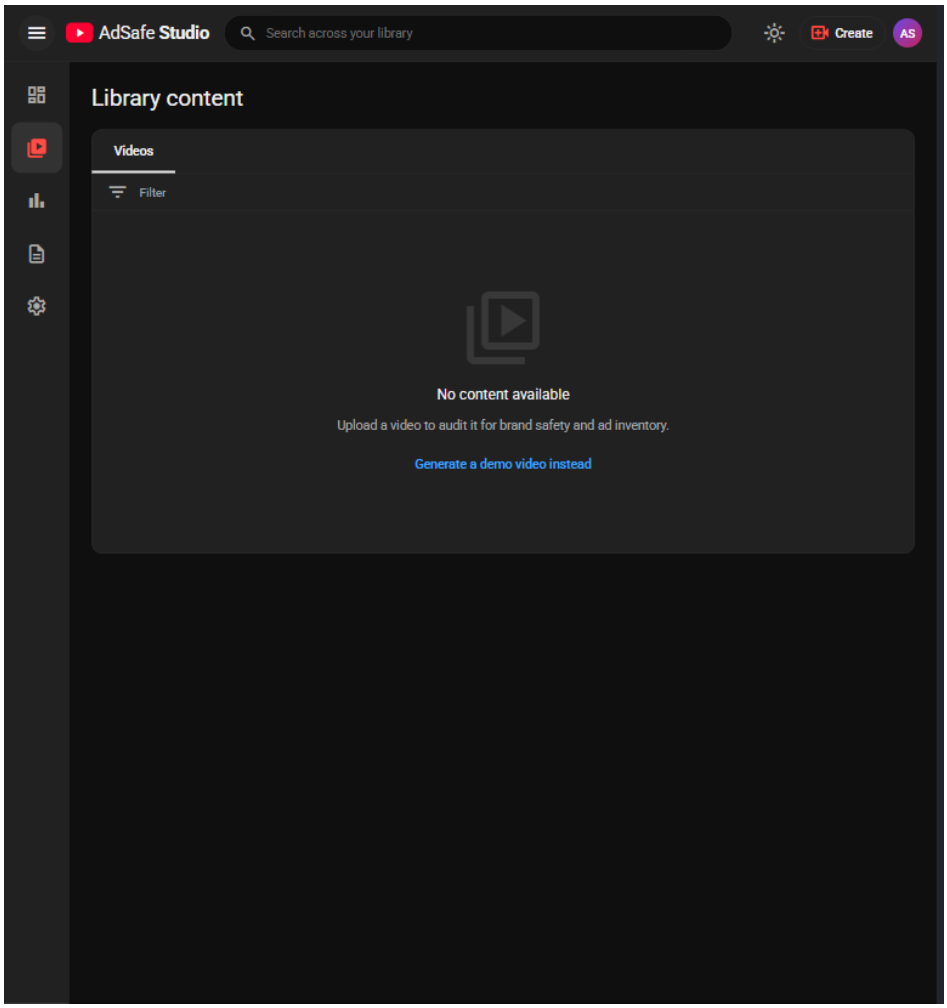


Figure 12: Laptop view — navigation collapsed to an icon rail

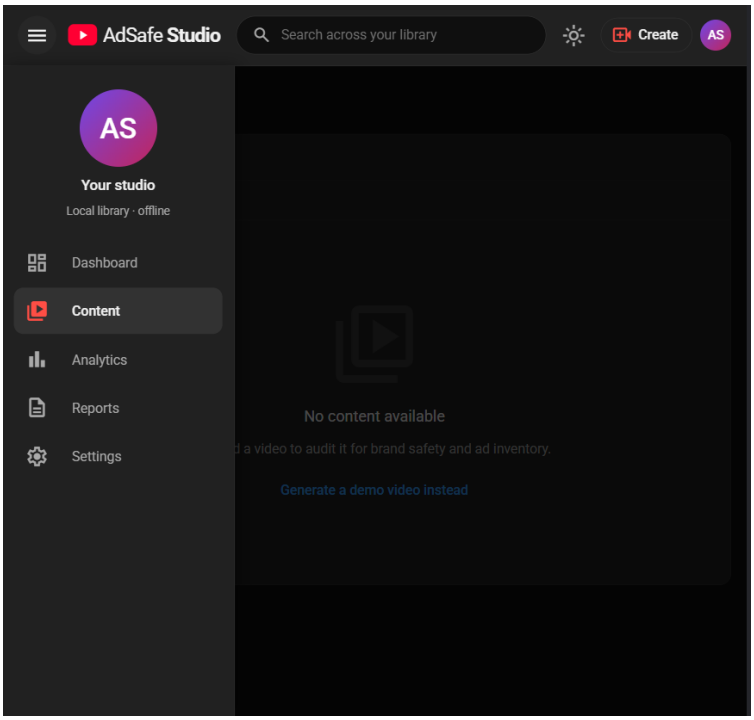


Figure 13: Tablet view — navigation presented as an overlay drawer

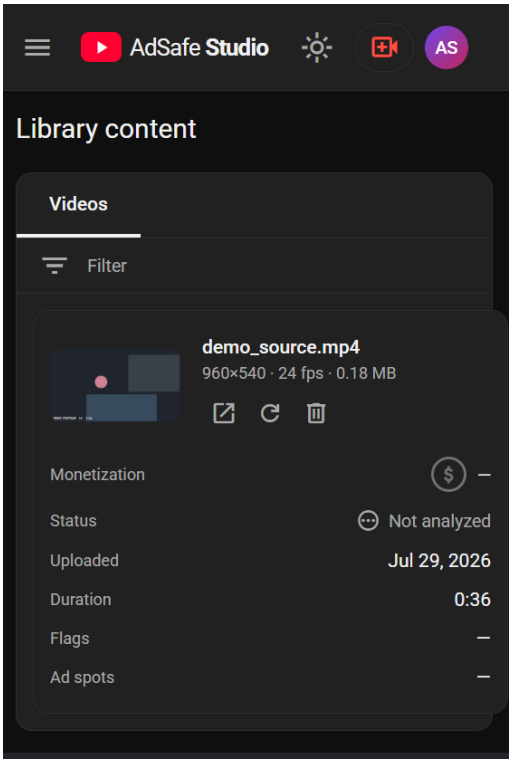


Figure 14: Mobile view — the content table re-rendered as labelled cards. A horizontally scrolling table is unusable at this width, so each row becomes a self-contained card with its column headings restated as field labels.

8. Usability Considerations

8.1 Consistency and Learnability

- A single persistent shell frames every screen, so navigation never relocates.
- Severity colours, badge shapes, and status vocabulary are identical on every screen where they appear.
- The layout follows established creator-tool conventions, reducing the learning required of the target user.

8.2 Efficiency

- Every primary destination is one click from any screen.
- Analysis runs in the background; the operator may navigate away and return.
- Timeline events seek the player directly, removing manual scrubbing — the single largest time cost in manual review.

- Reports are generated from the video being examined, without re-selecting it.

8.3 Feedback and Error Prevention

- Long operations report a named stage and a percentage, never an indeterminate spinner.
- Destructive actions require confirmation.
- Invalid input is identified against the field responsible, in plain language, and blocks submission rather than failing afterwards.
- Files are validated before upload begins, so a rejected file costs no transfer.
- Empty states explain the condition and offer the next action.

8.4 Accessibility

- Text contrast meets or exceeds WCAG 2.1 AA; interface components meet the 3:1 minimum. The action colour is themed specifically to satisfy this in dark mode.
- Colour is never the only signal — severity is always accompanied by a label.
- The status palette is verified for colour-blind separation.
- All controls are keyboard operable, including timeline events, and receive a visible focus indicator.
- Icon-only controls carry descriptive accessible names; status changes are announced to assistive technology.
- A closed navigation drawer is removed from the focus order.
- Motion is suppressed for users who request reduced motion.

8.5 Theming

Both a dark and a light theme are provided; the preference persists between sessions and is applied before first paint to avoid a flash of the wrong theme.

9. Prototype Screens

The high-fidelity prototype for this project is the **working application itself**. Rather than producing static mockups in a design tool, the interface was implemented directly and every figure in this document is a capture of the running system.

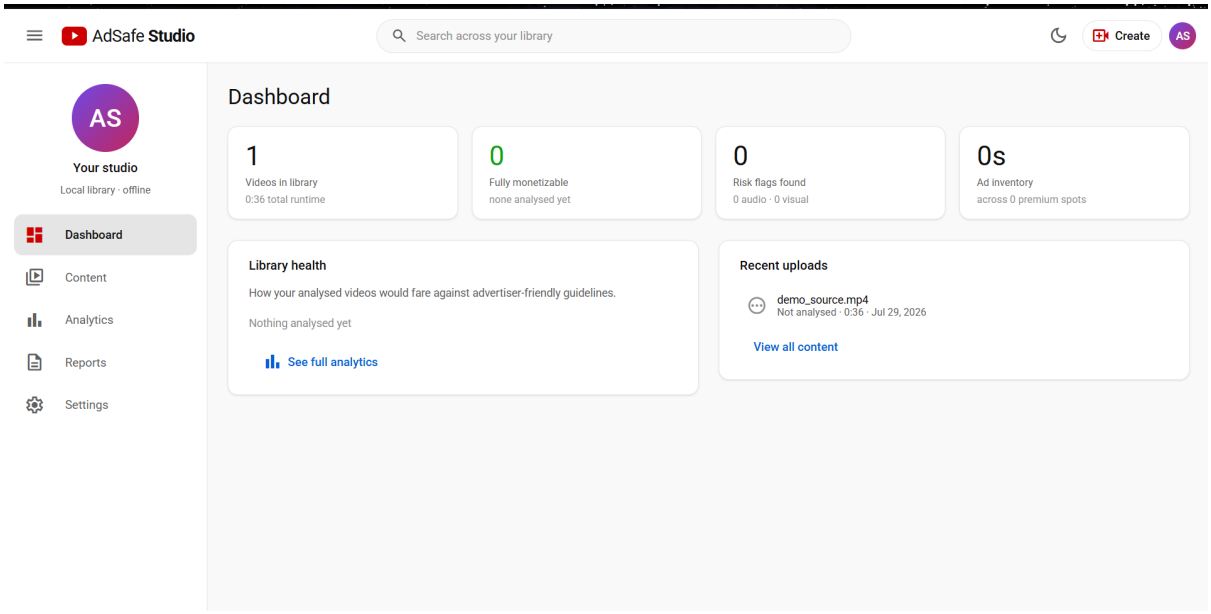


Figure 15: Light theme. Surfaces and text tokens invert; the action colour is substituted to maintain contrast.

Item	Detail
Implementation	React with Vite; custom design system in CSS
Interface served at	`http://localhost:8000`
Source repository	`https://github.com/meeeee21/autonomous-video-analyzer`
Prototype fidelity	Fully interactive; all navigation, validation, theming and responsive behaviour is live

The prototype exceeds what a design-tool mock would demonstrate: the navigation, validation, progress reporting, and responsive reflow shown in this document are genuine behaviour rather than linked static frames.

10. Conclusion

This User Interface Design Document defines the visual design, navigation structure, interaction behaviour, and validation rules of the Autonomous Video Ad-Placement & Contextual Safety Analyzer. Seven screens have been designed and implemented, spanning four responsive breakpoints and two themes, with colour and contrast decisions verified against measured accessibility criteria rather than chosen by appearance alone.

Because the interface is already implemented and operational, this document records a realised design rather than a proposal. It serves as the reference for the remaining implementation phases, during which the presentation layer stays fixed while the underlying detection engines are completed.

11. Submission Checklist

1. User Interface Design Document (PDF) — this document
2. Navigation Flow Diagram — Section 3
3. Minimum Six Screen Wireframes — Section 5, seven screens provided
4. High-Fidelity Prototype Screens — Section 5, live application captures
5. Prototype Link — Section 10, running application and source repository
6. Design Assets — colour palette and typography specifications, Section 4